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Introduction to Ethereum

Blockchain 2.0

Introduction

- Blockchain technology was first popularized with the launch of Bitcoin, which is considered the first generation of blockchain.
- Bitcoin demonstrated how blockchain could securely support a cryptocurrency system, but its primary purpose was limited to peer-to-peer digital cash transactions

Introduction

Although Bitcoin includes a small scripting language, it offers only very limited programmability and autonomy. Most transactions in Bitcoin need explicit triggers, and there is little room for complex automated processes.

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- In 2013, Vitalik Buterin, a young developer who had worked on Bitcoin, envisioned a more flexible blockchain platform. He introduced Ethereum, which became the second generation of blockchain.
- Unlike Bitcoin, Ethereum was designed not just as a cryptocurrency system but as a platform for decentralized applications. Its most important innovation was the concept of smart contracts.

Smart Contracts?

- Smart contracts are pieces of self-executing code stored on the blockchain.
- They allow transactions and agreements to be carried out automatically when certain conditions are met, without requiring human intervention.
- This autonomy makes it possible to support more complex and customizable decentralized applications (DApps).

Examples

- A loan repayment system that automatically releases collateral once full repayment is verified.
- A supply chain agreement where payment is triggered only when goods are delivered and verified.

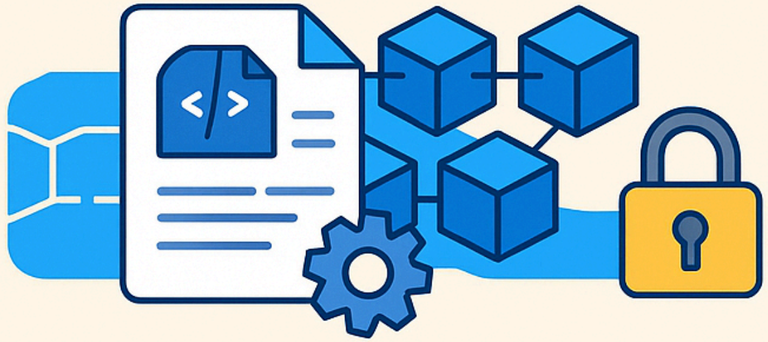
Summary

SMART CONTRACTS ON BLOCKCHAIN

SELF-EXECUTING CODE

CONDITION-BASED AUTOMATION

NO HUMAN INTERVENTION



LOAN REPAYMENT



REPAYMENT VERIFIED → RELESCE COLLATERAL





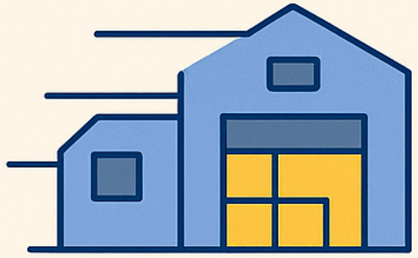
IF → THEN



SUPPLY CHAIN AGREEMENT



GOODS DELIVERED → TRIGGER PAYMENT





Ethereum as Blockchain Implementation

- It’s important to note that Ethereum is not a separate technology from blockchain—it is an implementation of blockchain technology, just like Microsoft Word is an implementation of word processing software
- Bitcoin was the first implementation; Ethereum followed as a more advanced one with broader functionality.

Ethereum as Blockchain Implementation

- While Bitcoin proved that digital currency could work, Ethereum showed that blockchain could be used for much more than cryptocurrency
- This shifted blockchain from a financial tool into a programmable infrastructure for many industries, including finance, healthcare, supply chains, and even voting systems.

Ethereum in Financial Service

- Ethereum has proven to be one of the strongest use cases of blockchain beyond cryptocurrency.
- Since financial services deal with transactions, trust, ownership, and intermediaries, Ethereum's features like transparency, provenance, disintermediation, and smart contracts make it a natural fit

Transparent Transaction & Provenance

- Transactions on Ethereum are publicly verifiable and not controlled by any single authority like a bank or credit card company.
- Every transfer—from one wallet address to another—is recorded immutably, allowing easy auditing.
- Provenance, or the ability to track ownership history, ensures assets (whether money, property, or goods) can be traced back to their origin.

Example

- In food supply chains, Ethereum could track produce (like romaine lettuce) from farm to table, enabling quick recalls during health outbreaks.

Disintermediation (Removing Middlemen)

- Traditional finance relies heavily on intermediaries (banks, brokers, lawyers). Each intermediary adds time, cost, and trust overhead.
- Ethereum eliminates many middlemen through decentralized trust, thereby reducing fees and speeding up settlements.

Automation with Smart Contracts

- In traditional finance, a human often validates conditions (e.g., loan approval, escrow release).
- Ethereum replaces this with smart contracts, which automatically execute when conditions are met.
- Example: If loan repayment is complete, collateral is instantly released.
- This reduces waiting time, lowers costs, and minimizes the chance of error or manipulation.

Currency & Payments

- Ethereum supports direct transfer of its native currency (Ether) between parties.
- Beyond Ether, Ethereum enables the creation of custom tokens, allowing businesses, startups, and even governments to design their own digital currencies or loyalty points.
- This makes cross-border remittances, micropayments, and peer-to-peer transfers faster and cheaper.

Fiat Vs Cryptocurrency

- Fiat currency (e.g., INR, USD, EUR) is backed by governments. Its value is defined by regulation and trust in the state.
- Cryptocurrency (e.g., Bitcoin, Ether) is not government-backed, and its value is determined by consensus in the market.
- Ethereum bridges both worlds:
- It allows cryptocurrency transfers.
- It supports tokenization of fiat-backed stablecoins (like USDT, USDC, or CBDCs in the future).

Real EState Transaction

- Real estate involves complex paperwork, multiple signatories (buyers, sellers, agents, lawyers), and escrow arrangements.
- With Ethereum, all rules can be codified into a smart contract:
- Buyer deposits money into escrow.
- Seller confirms property documents.
- Once all conditions are met, the contract automatically transfers title deeds to the buyer and funds to the seller.
- This reduces the need for lawyers, speeds up the process, and cuts costs dramatically.

Wider Implications

- Ethereum supports several financial applications often grouped under Decentralized Finance (DeFi):
- Lending & Borrowing – peer-to-peer loans without banks.
- Stablecoins – crypto assets pegged to fiat for price stability.
- Decentralized Exchanges (DEXs) – automated marketplaces for asset swaps.
- Insurance – payouts triggered by smart contracts when conditions (like flight delays) are met.

Key Benefits

- Transparency and trustless systems.
- Reduced costs via disintermediation.
- Efficiency and automation through smart contracts.
- Flexibility with custom tokens and DeFi applications.
- Broader asset coverage (not just money but also property, commodities, and even digital collectibles).

Summary

Ethereum in Financial Services



Transparent Transactions and Provenance



Automation wit Smart Contracts



Disintermediation



Fiat vs. Cryptocurrency



Fiat vs. Cryptocurrency



Wider Implications in Financial Services

- Key benefits
- Transparency
- Reduced costs

- Transparency
- Reduced costs
- Efficiency
- Flexibility

Ethereum in Digital Identity Management

- Identity refers to a unique attribute associated with a person, device, or entity.
- Traditionally, identity has been tied to official documents like passports, Aadhaar cards, or driver's licenses.
- In the digital world, however, identity is more fluid and often needs to be verified across platforms and devices.
- With the rise of the Internet of Things (IoT), not only people but also devices (e.g., smart refrigerators, doorbells, sensors) require unique digital identities to securely interact, share data, and execute automated instructions.

How Ethereum Manages?

- Ethereum provides a strong foundation for identity management through its account system and cryptography:
- Every Ethereum user creates an account consisting of a public-private key pair.
- The public key (address) acts as the digital identifier.
- The private key proves ownership and allows signing transactions.
- While this account proves control of the key, it doesn't inherently prove a real-world identity (e.g., Rajesh Kumar, or "this sensor is the kitchen temperature sensor").
- This gap is filled by identity frameworks and standards built on Ethereum

Identity Verification Via Claims & Standards

- Identity Verification via Claims and Standards (ERC-725)
- One important proposal is ERC-725, a standard for creating and managing decentralized identities on Ethereum.
- ERC-725 allows an Ethereum account to be associated with real-world claims.
- Claims are pieces of verified information linked to an identity.
- Example: A government agency can issue a claim that "Account X belongs to Michael Solomon."
- biometric claim (like a thumbprint) can also be stored and verified against the account.
- By collecting multiple claims, an Ethereum account gains assurance of authenticity, similar to presenting multiple identity documents in real life.

Advantages

- **Autonomy & Control: Users own their identity and decide what claims to share, unlike centralized systems controlled by governments or corporations.**
- **Security: Blockchain's immutability ensures identities and claims cannot be easily altered or forged.**
- **Interoperability: A blockchain-based identity can be recognized across platforms globally.**
- **Resilience: Even if a physical ID (like a driver's license) is lost, the digital identity remains verifiable through claims on the blockchain.**
- **Device Integration: IoT devices can use Ethereum identities to authenticate themselves and autonomously interact in a trusted environment.**

Application

- KYC & Regulatory Compliance: Banks or regulators can verify customer identities via blockchain claims, simplifying onboarding and audits.
- Cross-Border Identity: Migrants and international travelers can prove their identities digitally without relying solely on local documents.
- Decentralized IoT Security: Devices use Ethereum accounts to prove authenticity before exchanging data, reducing risks of spoofing or malicious access.
- Biometric Verification: Thumbprints, retina scans, or facial recognition can be securely linked as claims to blockchain accounts.

Summary

- Ethereum offers a decentralized, secure, and flexible framework for digital identity management. By combining cryptographic accounts with standards like ERC-725, Ethereum enables individuals, institutions, and devices to create trusted, verifiable, and autonomous digital identities. This not only enhances privacy and security but also enables seamless interactions in finance, IoT, governance, and beyond.

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Thank You For Your Attention